Lab 10 practice: double integrals and Green's theorem (Zill §9.10, 9.11, 9.12)

Prep for the Wed Nov 18 lab quiz. Closed book, about 10 minutes per problem. Problems 1 to 3 are iterated integrals (both orders, reversing the order, area and volume), problem 4 is polar coordinates, and problems 5 to 8 are Green's theorem. No figures are given: sketch each region yourself from the description, and write its limits before you integrate. Every answer was recomputed independently, and the Green's theorem problems were checked on both sides (`lab10_check.py`).

Iterated integrals. Type I region: $a \le x \le b,\ g_1(x) \le y \le g_2(x)$, and

$$\iint_R f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f\,dy\,dx$$

Type II region: $c \le y \le d,\ h_1(y) \le x \le h_2(y)$, and $\iint_R f\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f\,dx\,dy$. The inner limits may depend on the outer variable; the outer limits are constants. Area is $\iint_R dA$; the volume under $z = f \ge 0$ over R is $\iint_R f\,dA$. To reverse the order, sketch R, then read the limits the other way.

Polar coordinates. $x = r\cos\theta,\ y = r\sin\theta,\ x^2 + y^2 = r^2$, and

$$dA = r\,dr\,d\theta$$

The extra factor r is the Jacobian and is never optional. Circle $x^2 + y^2 = 2ax$ is $r = 2a\cos\theta$.

Green's theorem. If C is a piecewise smooth, simple, closed curve, positively oriented (counterclockwise, region on the left), R is the region it encloses, and P, Q have continuous first partials on an open region containing R and C, then

$$\oint_C P\,dx + Q\,dy = \iint_R \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\,dA$$

Clockwise traversal flips the sign. **Area:** $A = \frac{1}{2}\oint_C x\,dy - y\,dx$ (also $\oint x\,dy$ or $-\oint y\,dx$ alone).

Problem 1

Let R be the region bounded by the curves $y = x$ and $y = x^2$. Evaluate $\iint_R (x + 2y)\, dA$ as a Type I integral, then again as a Type II integral, and confirm the two agree.

Solution: the same region in both orders

The curves meet where $x = x^2$, that is at $(0,0)$ and $(1,1)$. For $0 < x < 1$ we have $x^2 < x$, so the parabola is the lower boundary and the line the upper one.

Type I. $0 \le x \le 1,\ x^2 \le y \le x$.

$$\int_0^1 \int_{x^2}^{x} (x + 2y)\, dy\, dx = \int_0^1 \Big[xy + y^2\Big]_{x^2}^{x} dx = \int_0^1 (2x^2 - x^3 - x^4)\, dx = \frac{2}{3} - \frac{1}{4} - \frac{1}{5} = \frac{13}{60}$$

Type II. Solve each curve for x: the line is $x = y$ and the parabola is $x = \sqrt{y}$. For $0 < y < 1$, $y < \sqrt{y}$, so $0 \le y \le 1,\ y \le x \le \sqrt{y}$.

$$\int_0^1 \int_{y}^{\sqrt{y}} (x + 2y)\, dx\, dy = \int_0^1 \Big[\frac{x^2}{2} + 2xy\Big]_{y}^{\sqrt{y}} dy = \int_0^1 \Big(\frac{y}{2} + 2y^{3/2} - \frac{5y^2}{2}\Big) dy = \frac{1}{4} + \frac{4}{5} - \frac{5}{6} = \frac{13}{60}$$

$$\boxed{\iint_R (x + 2y)\, dA = \frac{13}{60}}$$

Both orders agree, as they must. Here Type I was a little shorter; the point of the exercise is that either order works when both boundaries are single-valued functions.

Problem 2

Evaluate by reversing the order of integration.

(a) $\displaystyle\int_0^2 \int_{x/2}^1 e^{y^2}\, dy\, dx$

(b) $\displaystyle\int_0^1 \int_x^{2-x} y\, dy\, dx$, written first as a sum of Type II integrals.

Solution: when the given order cannot be integrated, or needs splitting

(a) $\int e^{y^2}\,dy$ has no elementary antiderivative, so integrating y first fails. The region is $0 \le x \le 2,\ x/2 \le y \le 1$: a triangle with vertices $(0,0)$, $(2,1)$, $(0,1)$, bounded by the line $y = x/2$ (so $x = 2y$), the line $y = 1$ and the y-axis. Reading it the other way, y runs from 0 to 1 and, at each y, x runs from the y-axis to the line, so $0 \le y \le 1,\ 0 \le x \le 2y$.

$$\int_0^1 \int_0^{2y} e^{y^2}\,dx\,dy = \int_0^1 2y\,e^{y^2}\,dy = \Big[e^{y^2}\Big]_0^1 = e - 1$$

The factor y from the inner integral is exactly what the substitution $u = y^2$ needs.

(b) The region is $0 \le x \le 1,\ x \le y \le 2 - x$: a triangle with vertices $(0,0)$, $(0,2)$, $(1,1)$, bounded by $y = x$ below, $y = 2 - x$ above and the y-axis on the left. The two slanted sides meet at $(1,1)$, and the right-hand boundary changes formula there. For $0 \le y \le 1$ the right edge is $x = y$; for $1 \le y \le 2$ it is $x = 2 - y$. So the region splits at $y = 1$:

$$\int_0^1 \int_0^{y} y\,dx\,dy + \int_1^2 \int_0^{2-y} y\,dx\,dy = \int_0^1 y^2\,dy + \int_1^2 y(2-y)\,dy = \frac{1}{3} + \Big[y^2 - \frac{y^3}{3}\Big]_1^2 = \frac{1}{3} + \frac{4}{3} - \frac{2}{3} = 1$$

Check in the original order: $\int_0^1 \frac{1}{2}\big[(2-x)^2 - x^2\big]dx = \int_0^1 (2 - 2x)\,dx = 1$.

$$\boxed{\text{(a) } e - 1, \qquad \text{(b) } 1}$$

In (b) the reversal was optional, since the original order was easy; it is there to show the split. A region whose right or left boundary changes formula needs one integral per formula.

Problem 3

(a) Find the area of the region R enclosed by the parabola $y = x^2$ and the line $y = 2x + 3$.

(b) Find the volume of the solid bounded above by the plane $z = 4 - y$, below by the xy-plane, and on the sides by the parabolic cylinder $y = x^2$.

Solution: area and volume as double integrals

(a) The curves meet where $x^2 = 2x + 3$, that is $(x-3)(x+1) = 0$, so at $(-1, 1)$ and $(3, 9)$. Between them the line is above the parabola (at $x = 0$: $3 > 0$). So $-1 \le x \le 3,\ x^2 \le y \le 2x + 3$.

$$A = \iint_R dA = \int_{-1}^{3} \int_{x^2}^{2x+3} dy\, dx = \int_{-1}^{3} (2x + 3 - x^2)\, dx = \Big[x^2 + 3x - \frac{x^3}{3}\Big]_{-1}^{3}$$

At $x = 3$: $9 + 9 - 9 = 9$. At $x = -1$: $1 - 3 + \frac{1}{3} = -\frac{5}{3}$. So $A = 9 + \frac{5}{3} = \frac{32}{3}$.

(b) The base region R is where the solid meets the plane $z = 0$. On that plane $z = 4 - y$ is zero at $y = 4$, and the cylinder $y = x^2$ meets $y = 4$ at $x = \pm 2$. So the base is $-2 \le x \le 2,\ x^2 \le y \le 4$, and on it $4 - y \ge 0$, so the plane really is the upper surface.

$$V = \iint_R (4 - y)\, dA = \int_{-2}^{2} \int_{x^2}^{4} (4 - y)\, dy\, dx = \int_{-2}^{2} \frac{(4 - x^2)^2}{2}\, dx$$

since $\int_{x^2}^{4} (4 - y)\, dy = \big[-\frac{(4-y)^2}{2}\big]_{x^2}^{4} = \frac{(4-x^2)^2}{2}$. Expand and use symmetry:

$$V = \int_0^2 (16 - 8x^2 + x^4)\, dx = 32 - \frac{64}{3} + \frac{32}{5} = \frac{256}{15}$$

$$\boxed{\text{(a) } A = \frac{32}{3}, \qquad \text{(b) } V = \frac{256}{15}}$$

Check (b) with the order reversed: $\int_0^4 \int_{-\sqrt{y}}^{\sqrt{y}} (4 - y)\, dx\, dy = \int_0^4 2\sqrt{y}\,(4 - y)\, dy = 2(\frac{64}{3} - \frac{64}{5}) = \frac{256}{15}$.

Problem 4

(a) Evaluate $\displaystyle\int_0^2 \int_0^{\sqrt{4-x^2}} e^{-(x^2+y^2)}\, dy\, dx$ by converting to polar coordinates.

(b) Evaluate $\displaystyle\iint_D x\, dA$, where D is the disk $x^2 + y^2 \leq 2x$.

(c) Find the volume of the solid between the paraboloid $z = 9 - x^2 - y^2$ and the xy-plane.

Solution: converting the region, the integrand, and the area element

(a) The region is $0 \le x \le 2,\ 0 \le y \le \sqrt{4-x^2}$: the quarter of the disk $x^2+y^2 \le 4$ in the first quadrant. In polar form that is $0 \le r \le 2,\ 0 \le \theta \le \pi/2$. The integrand becomes e^{-r^2} and $dA = r\,dr\,d\theta$.

$$\int_0^{\pi/2}\int_0^{2} e^{-r^2}\, r\,dr\,d\theta = \frac{\pi}{2}\Big[-\frac{1}{2}e^{-r^2}\Big]_0^{2} = \frac{\pi}{2}\cdot\frac{1-e^{-4}}{2} = \frac{\pi\,(1-e^{-4})}{4}$$

In rectangular coordinates the inner integral $\int e^{-y^2}dy$ is not elementary; the leftover r from dA is what lets the r-integral close.

(b) Complete the square: $x^2-2x+y^2 \le 0$ gives $(x-1)^2+y^2 \le 1$, the disk of radius 1 centred at $(1,0)$. In polar form, $r^2 \le 2r\cos\theta$, so $0 \le r \le 2\cos\theta$. The disk lies in $x \ge 0$, so $-\pi/2 \le \theta \le \pi/2$. The integrand is $x = r\cos\theta$.

$$\iint_D x\,dA = \int_{-\pi/2}^{\pi/2}\int_0^{2\cos\theta} r\cos\theta\cdot r\,dr\,d\theta = \int_{-\pi/2}^{\pi/2}\cos\theta\,\frac{8\cos^3\theta}{3}\,d\theta = \frac{8}{3}\int_{-\pi/2}^{\pi/2}\cos^4\theta\,d\theta$$

With $\int_{-\pi/2}^{\pi/2}\cos^4\theta\,d\theta = \frac{3\pi}{8}$ (use $\cos^2\theta = \frac{1+\cos 2\theta}{2}$ twice), the value is $\frac{8}{3}\cdot\frac{3\pi}{8} = \pi$. As a check, $x=1$ is the centroid of the disk and its area is π, so $\iint x\,dA = 1\cdot\pi$.

(c) The paraboloid meets $z=0$ on the circle $x^2+y^2=9$, so the base is the disk $0 \le r \le 3,\ 0 \le \theta \le 2\pi$, and $z = 9-r^2$ there.

$$V = \int_0^{2\pi}\int_0^{3}(9-r^2)\,r\,dr\,d\theta = 2\pi\Big[\frac{9r^2}{2}-\frac{r^4}{4}\Big]_0^{3} = 2\pi\Big(\frac{81}{2}-\frac{81}{4}\Big) = \frac{81\pi}{2}$$

$$\boxed{\text{(a) } \frac{\pi\,(1-e^{-4})}{4}, \qquad \text{(b) } \pi, \qquad \text{(c) } \frac{81\pi}{2}}$$

Problem 5

Verify Green's theorem for $\oint_C xy\,dx + x^2\,dy$, where C is the triangle with vertices $(0,0)$, $(1,0)$, $(1,2)$, traversed counterclockwise. Compute the line integral directly and the double integral, and compare.

Solution: both sides of the theorem

Hypotheses. C is a simple closed piecewise smooth curve (three straight segments), positively oriented as stated, and $P = xy$, $Q = x^2$ are polynomials, so their partials are continuous everywhere. Green's theorem applies.

Double integral. $Q_x = 2x$ and $P_y = x$, so $Q_x - P_y = x$. The triangle is bounded by $y = 0$, $x = 1$ and the line $y = 2x$ from $(0,0)$ to $(1,2)$, so $0 \le x \le 1,\ 0 \le y \le 2x$.

$$\iint_R x\,dA = \int_0^1 \int_0^{2x} x\,dy\,dx = \int_0^1 2x^2\,dx = \frac{2}{3}$$

Line integral. Counterclockwise order is $(0,0)$ to $(1,0)$ to $(1,2)$ and back to $(0,0)$.

- C_1: $(0,0)$ to $(1,0)$, $x = t,\ y = 0$, $0 \le t \le 1$, so $dy = 0$ and $xy = 0$. The integral is 0.
- C_2: $(1,0)$ to $(1,2)$, $x = 1,\ y = t$, $0 \le t \le 2$, so $dx = 0$ and $x^2\,dy = dt$. The integral is $\int_0^2 dt = 2$.
- C_3: $(1,2)$ to $(0,0)$, $x = t,\ y = 2t$, with t running from 1 down to 0. Then $dx = dt,\ dy = 2\,dt$ and $xy\,dx + x^2\,dy = 2t^2\,dt + 2t^2\,dt = 4t^2\,dt$, so the integral is $\int_1^0 4t^2\,dt = -\frac{4}{3}$.

$$\oint_C = 0 + 2 - \frac{4}{3} = \frac{2}{3}$$

$$\boxed{\oint_C xy\,dx + x^2\,dy = \iint_R x\,dA = \frac{2}{3}}$$

Both sides give $\frac{2}{3}$. On C_3 the parameter must run from 1 to 0 to follow the counterclockwise direction; using 0 to 1 would flip that piece to $+\frac{4}{3}$ and the sides would no longer agree.

Problem 6

Let C be the boundary of the region between the parabola $y = x^2$ and the line $y = 2x$, traversed counterclockwise. Use Green's theorem to evaluate

$$\oint_C -y^2\,dx + (xy + x)\,dy$$

Solution: a region bounded by two curves

Region. The curves meet where $x^2 = 2x$, at $(0,0)$ and $(2,4)$. Between them the line is above the parabola (at $x = 1$: $2 > 1$), so $0 \le x \le 2,\ x^2 \le y \le 2x$.

Hypotheses. C is made of two smooth arcs, it is simple and closed, and $P = -y^2$, $Q = xy + x$ are polynomials. The stated orientation is counterclockwise, which here means the parabola is traversed left to right and the line right to left (region on the left), so the theorem applies as written.

Integrand. $Q_x = y + 1$ and $P_y = -2y$, so $Q_x - P_y = 3y + 1$.

$$\iint_R (3y+1)\,dA = \int_0^2 \int_{x^2}^{2x} (3y+1)\,dy\,dx = \int_0^2 \Big[\frac{3y^2}{2} + y\Big]_{x^2}^{2x} dx = \int_0^2 \Big(5x^2 + 2x - \frac{3x^4}{2}\Big)dx$$

$$= \frac{40}{3} + 4 - \frac{48}{5} = \frac{116}{15}$$

$$\boxed{\oint_C -y^2\,dx + (xy+x)\,dy = \frac{116}{15}}$$

Direct check, two pieces. On the parabola $x = t,\ y = t^2$ with $0 \le t \le 2$: $\int_0^2 (t^4 + 2t^2)\,dt = \frac{176}{15}$. On the line $x = t,\ y = 2t$ with t from 2 down to 0: $\int_2^0 2t\,dt = -4$. The sum is $\frac{176}{15} - 4 = \frac{116}{15}$.

Problem 7

Use Green's area formula to find the area enclosed by the astroid

$$x = 2\cos^3 t, \qquad y = 2\sin^3 t, \qquad 0 \le t \le 2\pi$$

Solution: area as a line integral

The curve is simple and closed ($t = 0$ and $t = 2\pi$ give the same point $(2, 0)$) and is smooth except at its four cusps, so it is piecewise smooth. As t increases from 0 it moves from $(2, 0)$ toward $(0, 2)$, so it is traversed counterclockwise and the area formula $A = \frac{1}{2}\oint_C x\,dy - y\,dx$ applies without a sign change.

$$dx = -6\cos^2 t \sin t\,dt, \qquad dy = 6\sin^2 t \cos t\,dt$$

$$x\,dy - y\,dx = (12\cos^4 t \sin^2 t + 12\sin^4 t \cos^2 t)\,dt = 12\sin^2 t\cos^2 t\,(\cos^2 t + \sin^2 t)\,dt = 3\sin^2 2t\,dt$$

using $\sin t \cos t = \frac{1}{2}\sin 2t$. Then

$$A = \frac{1}{2}\int_0^{2\pi} 3\sin^2 2t\,dt = \frac{3}{2}\int_0^{2\pi} \frac{1-\cos 4t}{2}\,dt = \frac{3}{2}\cdot\pi = \frac{3\pi}{2}$$

$$\boxed{A = \frac{3\pi}{2}}$$

For general half-width a the same computation gives $\frac{3\pi a^2}{8}$; here $a = 2$ gives $\frac{3\pi}{2}$. Compare with the circle of radius 2 (area 4π): the astroid is much thinner, which is the right order of magnitude.

Problem 8

(a) Let C be the circle $x^2 + y^2 = 4$ traversed **clockwise**. Evaluate $\oint_C -y^3\,dx + x^3\,dy$.

(b) Let C be the unit circle $x^2 + y^2 = 1$ traversed counterclockwise, and $\oint_C \dfrac{-y\,dx + x\,dy}{x^2 + y^2}$. Show that Green's theorem cannot be used, and find the value another way.

Solution: reversing an orientation, and a failed hypothesis

(a) Orientation. Hypotheses: C is a simple closed smooth curve and $P = -y^3,\ Q = x^3$ are polynomials. But Green's theorem as stated needs positive orientation, and this C is clockwise. Let $-C$ denote the same circle counterclockwise. Reversing direction changes the sign of a line integral, so

$$\oint_C = -\oint_{-C}$$

Apply the theorem to $-C$. $Q_x - P_y = 3x^2 + 3y^2 = 3r^2$ on the disk $0 \le r \le 2$:

$$\oint_{-C} = \iint 3r^2 \cdot r\, dr\, d\theta = \int_0^{2\pi} \int_0^2 3r^3\, dr\, d\theta = 2\pi \cdot \frac{3 \cdot 16}{4} = 24\pi$$

$$\boxed{\text{(a) } \oint_C -y^3\, dx + x^3\, dy = -24\pi}$$

(b) Hypothesis check. Here $P = \dfrac{-y}{x^2+y^2}$ and $Q = \dfrac{x}{x^2+y^2}$ are not defined at the origin, and the origin lies inside the disk bounded by C. There is no open region containing the closed disk on which P and Q have continuous partials, so Green's theorem does not apply.

The failure is real, not a technicality. Away from the origin $Q_x = \dfrac{y^2 - x^2}{(x^2+y^2)^2} = P_y$, so a careless use of the theorem would give $\iint 0\, dA = 0$. Compute the line integral directly instead. With $x = \cos t,\ y = \sin t$, $0 \le t \le 2\pi$ (counterclockwise), and $x^2 + y^2 = 1$ on the curve:

$$\oint_C \frac{-y\, dx + x\, dy}{x^2+y^2} = \int_0^{2\pi} (\sin^2 t + \cos^2 t)\, dt = 2\pi$$

$$\boxed{\text{(b) Green's theorem fails (singularity at the origin); the integral is } 2\pi \neq 0}$$

The value 2π is the total change in the polar angle around the origin. For a closed curve that does not enclose the origin, for instance the circle of radius 1 centred at $(3, 0)$, the hypotheses do hold and the integral is 0.